

1 Top risky paths

1.1 CTSM

1. Restart → get_curr_date → chkrc
2. Restart → init_vegrootfr
3. Restart → get_proc_global
4. InitializeRealize → initialize2 → hist_addfld1d → allhistfldlist_addfld → get_proc_global
5. InitializeRealize → initialize2 → hist_addfld1d → allhistfldlist_addfld → get_proc_bounds
6. Restart → CNVegMatrixRest
7. Restart → CNSoilMatrixRest
8. lnd_init → initialize2 → hist_addfld1d → allhistfldlist_addfld → get_proc_global
9. InitializeRealize → initialize2 → hist_addfld1d → allhistfldlist_make_active
10. lnd_init → initialize2 → hist_addfld1d → allhistfldlist_addfld → get_proc_bounds

1.2 CWatM

1. writenetcdf → cbinding
2. writenetcdf → getmeta
3. groundwater_modflow.initial → decompress
4. loadsetclone → loadmap → mapattrNetCDF → readCoord → readCoordNetCDF
5. groundwater_modflow.initial → is_float
6. loadsetclone → loadmap → cbinding
7. readnetcdf2 → mapattrNetCDF → readCoord → readCoordNetCDF
8. readnetcdf2 → cbinding
9. writeIniNetcdf → cbinding
10. loadsetclone → loadmap → compressArray

1.3 DBH

1. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data → Interpolate_tm → LOTPS → LOCAL → SGECD → SGEFA → SAXPY
2. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data → Interpolate_tm → LOTPS → LOCAL → SGECD → SGEFA → SSCAL
3. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data → Interpolate_n_summ → LOTPS → LOCAL → SGECD → SGEFA → SAXPY
4. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data → Interpolate_tmax → LOTPS → LOCAL → SGECD → SGEFA → SAXPY

5. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data
→ Interpolate_tmin → LOTPS → LOCAL → SGECO → SGEFA → SAXPY
6. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data
→ Interpolate_um → LOTPS → LOCAL → SGECO → SGEFA → SAXPY
7. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data
→ Interpolate_fsm → LOTPS → LOCAL → SGECO → SGEFA → SAXPY
8. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data
→ Interpolate_rsum → LOTPS → LOCAL → SGECO → SGEFA → SAXPY
9. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data
→ Interpolate_sun → LOTPS → LOCAL → SGECO → SGEFA → SAXPY
10. (main:ETPTREND.f) → ANA_ReferET_TREND → Get_potential_ET → Get_Grid_ATM → Read_Interpolate_data
→ Interpolate_n_summ → LOTPS → LOCAL → SGECO → SGEFA → SSCAL

1.4 GR4J

1. get_versions → versions_from_vcs → run_command
2. get_versions → versions_from_expanded_variables
3. gr4j → s_curves2
4. gr4j → s_curves1
5. get_versions → versions_from_parentdir

1.5 H08

1. main_human → wrte_bints3 → wrte_bints2 → wrte_binary
2. main_ftcs → wrte_bints3 → wrte_bints2 → wrte_binary
3. main_human → read_result → read_binary
4. main_ftcs → read_result → read_binary
5. main_human → wrte_bints2 → wrte_binary
6. main_ftcs → wrte_bints2 → wrte_binary
7. main_human → calc_damdem → read_binary
8. main_human → read_binary
9. main_ftcs → calc_damdem → read_binary
10. main_ftcs → read_binary

1.6 HBV

1. nse_cost → hbv_main

1.7 HydroPy

1. `get_daily_cover` → `log_cells.add_value`
2. `variables.write2stream` → `output_stream.write_variable`
3. `get_daily_cover` → `monthly_interpol`
4. `variables.add_variable` → `output_stream.var2stream`
5. `forcing.__init__` → `dataset2double`
6. `parameter.get_flow_properties` → `eval_flowfield`
7. `variables.initialize` → `dataset2double`
8. `get_potential_snowmelt` → `daylength`
9. `get_potential_snowmelt` → `log_cells.add_value`
10. `update_skincanopy` → `log_cells.add_value`

1.8 HYPE

1. HYSS → `linesearch_methods_calibration` → `new_Brent_method` → `linesearch_HYSS` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `particle_processes_for_runoff` → `retention_pool`
2. HYSS → `linesearch_methods_calibration` → `quasiNewton_algorithm` → `direction_multiplier` → `linesearch_HYSS` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `particle_processes_for_runoff` → `retention_pool`
3. HYSS → `linesearch_methods_calibration` → `new_Brent_method` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `particle_processes_for_runoff` → `retention_pool`
4. HYSS → `linesearch_methods_calibration` → `quasiNewton_algorithm` → `grad_criterium_function` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `particle_processes_for_runoff` → `retention_pool`
5. HYSS → `linesearch_methods_calibration` → `new_Brent_method` → `linesearch_HYSS` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `add_dry_deposition_to_landclass` → `atmdep_in_loss` → `production_pool`
6. HYSS → `linesearch_methods_calibration` → `quasiNewton_algorithm` → `direction_multiplier` → `linesearch_HYSS` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `add_dry_deposition_to_landclass` → `atmdep_in_loss` → `production_pool`
7. HYSS → `linesearch_methods_calibration` → `quasiNewton_algorithm` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `particle_processes_for_runoff` → `retention_pool`
8. HYSS → `linesearch_methods_calibration` → `new_Brent_method` → `linesearch_HYSS` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `add_dry_deposition_to_landclass` → `production_pool`
9. HYSS → `stage_MonteCarlo` → `stage_MonteCarlo_core_function` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `particle_processes_for_runoff` → `retention_pool`
10. HYSS → `linesearch_methods_calibration` → `quasiNewton_algorithm` → `direction_multiplier` → `linesearch_HYSS` → `function_to_minim` → `run_model_crit` → `model` → `soilmodel_0` → `add_dry_deposition_to_landclass` → `production_pool`

1.9 MHM

1. `run → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → Genuchten → message`
2. `run → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → hydro_cond → message`
3. `run_or_optimize → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → Genuchten → message`
4. `mhm_driver → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → Genuchten → message`
5. `run_or_optimize → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → hydro_cond → message`
6. `mhm_driver → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → hydro_cond → message`
7. `run → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_SMhorizons → message`
8. `run_or_optimize → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_SMhorizons → message`
9. `mhm_driver → mhm_interface_run → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_SMhorizons → message`
10. `run_with_parameter → mhm_eval → mhm_interface_run_prepare → mpr_eval → mpr → mpr_sm → Genuchten → message`

1.10 ORCHIDEE

1. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → calc_Nlimfert → ipslerr_p`
2. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → grazing_intake_complementation → Euler_funct`
3. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → balance_energy_cow → Euler_funct`
4. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → Milk_Animal_cow → Euler_funct`
5. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → Milk_Animal_cow_d → Euler_funct`
6. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → balance_energy_calf → Euler_funct`
7. `driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → balance_energy_heifer → Euler_funct`

8. driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main_dynamic → grazing_intake_complementation → auto_complementation_dairy → calcul_NEI_cow_d
9. driver → intersurf_main_2d → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → Animaux_main → Grazing_intake → Euler_funct
10. orchidedriver → sechiba_main → slowproc_main → stomate_main → StomateLpj → main_grassland_management → calc_N_limfert → ipslerr_p

1.11 PCR-GLOBWB

1. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjClone → singleTryNetcdf2PCRobjClone → regridData2FinerGrid
2. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjClone → singleTryNetcdf2PCRobjClone → getMapAttributesALL
3. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjClone → singleTryNetcdf2PCRobjClone → findLastYearInNCTime
4. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjClone → singleTryNetcdf2PCRobjClone → findFirstYearInNCTime
5. singleTryNetcdf2PCRobjClone_version.until.2020.07.14 → regridData2FinerGrid
6. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjCloneWithoutTime → singleTryNetcdf2PCRobjCloneWithoutTime → regridData2FinerGrid
7. singleTryNetcdf2PCRobjClone_version.until.2020.07.14 → getMapAttributesALL
8. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjCloneWithoutTime → singleTryNetcdf2PCRobjCloneWithoutTime → getMapAttributesALL
9. readPCRmapClone → singleTryReadPCRmapClone → netcdf2PCRobjCloneWithoutTime → singleTryNetcdf2PCRobjCloneWithoutTime → getFullPath
10. singleTryNetcdf2PCRobjClone_version.until.2020.07.14 → findLastYearInNCTime

1.12 SACRAMENTO

1. EXSAC → SAC1 → FGFR1
2. sac_initialize → initialize.from.file → read_sac_parameters
3. sac_initialize → initialize.from.file → read_sac_statefiles → unix_to_datehr → calendar_date
4. sac_update → advance.in.time → unix_to_datehr → calendar_date
5. parse → split → compact
6. sac_update → advance.in.time → solve_sac → write_sac_output
7. multi_driver → unix_to_datehr → calendar_date
8. sac_update → advance.in.time → solve_sac → read_areal_forcing
9. date_diff_ndays → julianday_scalar

10. sac_update → advance_in_time → solve_sac → write_sac_statefile

1.13 SWAT

1. calsoft_control → calsoft_hyd → time_control → command → hru_control → wetland_control
→ conditions → cond_real
2. calsoft_control → calsoft_hyd → time_control → command → gwflow_simulate → gwflow_rech
3. calsoft_control → calsoft_hyd → time_control → command → hru_control → actions → cn2_init
→ curno → ascrv
4. calsoft_control → caltsoft_hyd → time_control → command → hru_control → wetland_control
→ conditions → cond_real
5. calsoft_control → calsoft_hyd → time_control → command → hru_control → actions → pl_burnop
→ curno → ascrv
6. calsoft_control → caltsoft_hyd → time_control → command → gwflow_simulate → gwflow_rech
7. calsoft_control → caltsoft_hyd → time_control → command → hru_control → actions → cn2_init
→ curno → ascrv
8. calsoft_control → caltsoft_hyd → time_control → command → hru_control → actions → pl_burnop
→ curno → ascrv
9. calsoft_control → calsoft_hyd → time_control → command → hru_control → conditions → cond_real
10. calsoft_control → calsoft_hyd → time_control → command → hru_control → actions → curno
→ ascrv

2 Non risky paths (random sample from bottom 20%)

2.1 CTSM

1. `glc.timer.print.all` → `glc_timer_print` → `exit_glc` → `exit_message_environment`
2. `POP` → `output_driver` → `write_history` → `glc.io.write_history`
3. `init_grid1` → `flushm`
4. `initGridcells` → `set_landunit_ice` → `subgrid.get_info_glacier_mec`
5. `is_end_curr_year` → `get_curr_date` → `chkrc`

2.2 CWatM

1. `mapattrNetCDFMeteo` → `readCoordNetCDF`
2. `mainwarm` → `CWATMex2` → `ModelFrame.run`
3. `mainwarm` → `headerinfo`
4. `write_new_excel` → `open_workbook`
5. `test_cwatm` → `cwatm`

2.3 DBH

1. `Get_NDVI` → `Get_Ancil_latlon` → `Patch_Tiled_file` → `ConvI2C`
2. `(main:VD.f)` → `ANA_LONGWV_TREND` → `Get_Grid_ATM` → `Read_GEO_hrd`
3. `(main:LAITREND.f)` → `ANA_LAI_TREND` → `Get_LAI` → `Read_GEO_hrd`
4. `SIB_DHM` → `Downscale.swdown` → `sunset`
5. `SIB_DHM` → `SIB2_grid` → `Read_SIB_status`

2.4 GR4J

1. `get_versions` → `versions_from_parentdir`

2.5 H08

1. `htlinear` → `conv_r2tor1`
2. `prog_hlf2one` → `conv_r1tor2`
3. `prog_map_C05` → `conv_r2tor1`
4. `prog_frcgw` → `wrte_ascii2`
5. `htuscale` → `conv_r2tor1`

2.6 HBV

1. `nse_cost` → `hbv_main`

2.7 HydroPy

1. `get_skinevap` → `monthly_interpol`
2. `update_surface_water` → `correct_neg_stor`
3. `parameter.get_parameter` → `dataset2double`
4. `update_riverflow` → `routing_cascade` → `linear_cascade`
5. `update_soil` → `correct_neg_stor`

2.8 HYPE

1. `HYSS` → `DEMC_simulation` → `DEMC_runmedianparameters` → `run_model_simout` → `prepare_outputfiles` → `open_timefiles_classes` → `write_integer_header`
2. `HYSS` → `DEMC_simulation` → `run_model_simout` → `save_mapfiles` → `create_filename_for_variable` → `scalar_upper_case`
3. `valid_snowmelt_options` → `check_input` → `add_testcase`
4. `write_subbasinfiles` → `compute_basinoutput` → `check_and_acc_one_data` → `acc_one_data`
5. `HYSS` → `DEMC_simulation` → `DEMC_runmedianparameters` → `run_model_simout` → `set_optim_modpar`

2.9 MHM

1. `find_nearest` → `find_nearest_idx`
2. `FortranFile.write` → `Routine.to_str`
3. `ud` → `u2d` → `j2d`
4. `read_LCover` → `open_new_nml`
5. `read_percolation1` → `open_new_nml`

2.10 ORCHIDEE

1. `scatter_omp_l2` → `orch_scatter_omp_lgen` → `barrier2_omp`
2. `init_intersurf` → `barrier2_omp`
3. `grid_toij_2d` → `latlon_to_ij` → `llij_ps_wgs84`
4. `gather_omp_i0` → `orch_gather_omp_simple_igen` → `print_omp_function` → `barrier2_omp`
5. `forcing_read` → `grid_stuff` → `grid_topolylist` → `haversine_regxytoploy` → `ij_to_latlon` → `ijll_albers_nad83`

2.11 PCR-GLOBWB

1. `ShortwaveRadiation.update` → `compute_radsw_ext`
2. `HamonPotET` → `dayLength`
3. `ShortwaveRadiation.parameter_initialization` → `get_patm_cor`

4. shortwave_radiation.parameter_initialization → get_tau_a
5. ScaledBmiPCRGlobWB.set_value → BmiPCRGlobWB.set_value

2.12 SACRAMENTO

1. insertstr → shiftstr
2. date_to_unix → parse_date
3. value_si → value_dr
4. sac_finalize → cleanup
5. parse → removebksl

2.13 SWAT

1. calsoft_control → calsoft_hyd_bfr → calsoft_hyd_bfr_perc → time_control → command → ch_salt_output
2. calhard_control → time_control → basin_sw_init
3. calsoft_control → calsoft_hyd_bfr → calsoft_hyd_bfr_surq → time_control → command → wet_salt_output
4. calsoft_control → calsoft_hyd_bfr → calsoft_hyd_bfr_perc → time_control → command → water_allocation_output
5. calhard_control → time_control → command → res_salt_output